

Nayeema Nonta

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PhD student specializing in LLM safety and efficient training, with a KDD oral, NeurIPS publication, and research experience in LLM alignment at NRC Canada.

RELEVANT EXPERIENCE

PhD Research Intern – LLM Safety & Alignment

Sep 2026 – Present

National Research Council Canada

- Investigating how internal representations of harm, truthfulness, and user state evolve across multi-turn LLM conversations and designing interventions to improve model safety and robustness.

Graduate Researcher – LLM Safety & Efficient Training

Dec 2024 – Present

University of Waterloo, Critical ML Lab

- Designed and implemented geometry-constrained LLM fine-tuning methods in PyTorch/Transformers to mitigate safety degradation during adaptation, building PEFT training and safety–utility evaluation pipelines across Llama and Qwen models; earlier versions were presented as **two ICLR 2026 workshop papers**.
- Contributed to TamperBench, a framework for stress-testing LLM safety, by implementing a refusal-ablation attack and adding support for activation-based fine-tuning methods; the resulting work was published as a **KDD 2026 oral paper**.
- Implemented projection-aware and recovery-scaling optimizer components and derived time-complexity bounds for a full-parameter LLM training method, **reducing training time by up to 65% vs. GaLore (NeurIPS 2025)**.

Research Assistant – Network Optimization

Jan 2024 – May 2025

University of Waterloo, Prof. Fatma Gzara

- Developed exact algorithms for an **NP-hard constrained shortest-path problem** on DAGs, cyclic graphs, and SINR-based maps, optimizing UAV path cost while maintaining minimum signal strength to base towers.
- Benchmarked the algorithms against a MILP formulation solved with Gurobi, achieving a **17.87× average speedup** and up to **206.33× faster runtime** on highly constrained large instances.

Machine Learning Software Developer Intern

Sep 2023 – Dec 2023

Core Avionics & Industrial Inc. (CoreAVI)

- Developed **four computer-vision applications in C** for CoreAVI's Vulkan SC-based safety-critical AI library, including from-scratch implementations of MobileNetV2 and an RBF-kernel SVM, plus FFT-based image processing.
- Built Python tooling to convert PyTorch/torchvision models through ONNX to NNEF for **deterministic inference in a safety-critical runtime**, including preprocessing and input-validation pipelines for model compatibility.

Rapid App Development Intern

Jan 2023 – Apr 2023

Nuclear Promise X (NPX)

- Built Power Apps to digitize paper-based time, observation, and status logging for project management teams and site workers at Bruce Power and Ontario Power Generation (OPG).
- Implemented SQL stored procedures and views, and built Azure DevOps **CI/CD pipelines** to support app delivery.

Business Systems Intern

Sep 2021 – Dec 2021

Nissan Canada Finance

- Coordinated cross-functional Salesforce CRM delivery, executing **150+ UAT/SIT tests** and managing business-critical JIRA issues to validate functional requirements and release readiness.

SELECTED PEER-REVIEWED PUBLICATIONS

[Oral] **TamperBench: Systematically Stress-Testing LLM Safety Under Fine-Tuning and Tampering**

Saad Hossain, Tom Tseng, Punya Syon Pandey, Samanvay Vajpayee, Matthew Kowal, Nayeema Nonta, Samuel Simko, Stephen Casper, Zhijing Jin, Kellin Pelrine, Sirisha Rambhatla

32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining, 2026.

[Poster] **Refusal-Orthogonal Gated Editing for Safer Localized LLM Adaptation**

Nayeema Nonta, Sirisha Rambhatla

14th International Conference on Learning Representations (ICLR), 2026 — 3rd Workshop on Test-Time Updates.

[Poster] **Safe Downstream Adaptation of LLMs via Refusal-Orthogonal Gated Editing**

Nayeema Nonta, Sirisha Rambhatla

14th International Conference on Learning Representations (ICLR), 2026 — Workshop on Representational Alignment.

[Poster] Random but Right: A Geometric Explanation for Efficient LLM Training

Sahar Rajabi, Nayeema Nonta, Sirisha Rambhatla

14th International Conference on Learning Representations (ICLR), 2026 — Workshop on Geometry-grounded Representation Learning and Generative Modeling (GRAM).

[Poster] Principled Randomized Exploration of Gradient Subspaces for Efficient LLM Training

Sahar Rajabi, Nayeema Nonta, Sirisha Rambhatla

14th International Conference on Learning Representations (ICLR), 2026 — Workshop on Deep Generative Models in Machine Learning.

[Poster] SubTrack++: Gradient Subspace Tracking for Scalable LLM Training

Sahar Rajabi, Nayeema Nonta, Sirisha Rambhatla

39th Conference on Neural Information Processing Systems (NeurIPS), 2025.

PROJECTS

Electric Plane Data Visualization & Predictive Analytics

May 2023 – Apr 2024

Award-winning University of Waterloo capstone | Python, scikit-learn, PostgreSQL

- Developed an **award-winning predictive analytics system** with the Waterloo Institute for Sustainable Aeronautics to analyze and forecast the effects of Canadian weather on electric-aircraft performance and battery consumption.
- Engineered a **PostgreSQL ETL pipeline** to spatiotemporally align large-scale flight telemetry with local METAR weather observations, creating analysis-ready datasets for predictive modeling of weather-driven aircraft performance.

EDUCATION

PhD Student, Management Science and Engineering

May 2025 – Present

University of Waterloo

Direct MASc-to-PhD transfer; master's degree not conferred. Advisor: Prof. Sirisha Rambhatla.

BASc, Management Engineering (Honours, Co-op), Computing Option, Distinction

Sep 2019 – Apr 2024

University of Waterloo

TECHNICAL SKILLS

Programming Languages | Python, C, SQL, Bash

Certifications | AWS Certified Solutions Architect – Associate (ID: 323c2b3ab4aa4d5c80a7a0251ecf7cb3)

Libraries & Tools | NumPy, PyTorch, Scikit-Learn, Gurobi, Transformers, PEFT/LoRA, vLLM, Slurm, Git, Codex/Claude Code

AWARDS

President's Scholarship (2019).

Gregory Zhang e-Zinc Award for Capstone Design (2024).

Management Engineering Design Award (2024).

Engineering Dean's Entrance Award (2025).

TALKS & PRESENTATIONS

Poster & Spotlight Talk

Mar 2026

2nd Annual Cybersecurity and Privacy Institute Graduate Student Conference

Safety-Aware Gated Editing (SAGE): Refusal-Orthogonal Activation Editing for Safe Low-Resource Adaptation.

SERVICE, LEADERSHIP, & OTHER EXPERIENCE

Reviewer – ICLR 2026 Workshops

2026

- Reviewed for Scaling Post-training for LLMs (SPOT) and Catch, Adapt, Operate (CAO): Monitoring Models Under Drift.

Teaching Assistant – University of Waterloo

Sep 2025 – Aug 2026

- Designed/Led tutorials and provided constructive feedback across four courses in engineering design, work design, applications of management engineering, and decision support systems (MSE 302, 131, 433, and 436).

Equity, Diversity, and Inclusion (EDI) Director – Women in Engineering (WiE)

2025 – Present

- Coordinated communications and logistics for **14 industry partners** at the WiE Graduate Career Fair.
- Coordinated a panel and period-product drive with local Canadian charity Help a Girl Out (HAGO)

Community Impact Initiatives Leader – Nuclear Promise X

2023

- Led a fundraising campaign and charity auction that raised **\$6,200** for Turkey–Syria earthquake relief.

Engineering Ambassador & Orientation Leader/Director – University of Waterloo

2019 – 2023

- Managed a team of **10 leaders** organizing events for **1,800+ first-year students**.